

SCALING UP TRAINING CONTINUOUS LATENT REASONING: ON THE POWER OF SOFT THINKING AND REINFORCEMENT LEARNING

Zheheng Zhang

Zhenhai High School
Zhenhai, Ningbo, Zhejiang, China, 315200
elliott.zh@proton.me

Chengyu Wu

Zhenhai High School
Zhenhai, Ningbo, Zhejiang, China, 315200
7086cmd@gmail.com

Mingshu Chen

Fudan University
Shanghai, China, 200433

Chengyang Zhu

Fudan University
Shanghai, China, 200433

ABSTRACT

We proposed a novel paradigm for empowering reasoning LLMs by combining Soft Thinking, a variant of continuous reasoning, with Large Scale Reinforcement Learning. Unlike previous methods of with-training continuous reasoning, our approach directly demonstrates state-of-the-art performance on difficult reasoning-required benchmarks such as AIME and MATH-500, showcasing significant stability and full potential to scaling up. Our code is available at GitHub and benchmark dataset available at HuggingFace.